

**Multiple Choice:**

1. What is the purpose of a compiler in programming?
 - A. To convert source code into machine code
 - B. To execute program instructions
 - C. To debug and test programs
 - D. To manage memory allocation
2. What is the role of an IDE (Integrated Development Environment)?
 - A. To write documentation for programs
 - B. To translate code into a different programming language
 - C. To provide tools for software development
 - D. To execute and run programs
3. What is a variable in programming?
 - A. A constant value that cannot be changed
 - B. A function that returns a value
 - C. A data type that stores characters
 - D. A named storage location that holds data
4. What is the purpose of a conditional statement in programming?
 - A. To repeat a block of code multiple times
 - B. To declare and define functions
 - C. To control the flow of execution based on a condition
 - D. To assign values to variables
5. What does the term "algorithm" refer to in programming?
 - A. A type of programming language
 - B. A set of rules or steps to solve a problem
 - C. A software testing technique
 - D. A programming error or bug
6. What is the purpose of the "while" loop in programming?
 - A. To execute a block of code a specified number of times
 - B. To define a function in the program
 - C. To repeat a block of code until a condition is no longer true
 - D. To assign values to variables
7. What is a function in programming?
 - A. A group of related data items
 - B. A reserved keyword in a programming language
 - C. A block of reusable code that performs a specific task
 - D. A built-in data type in programming languages
8. What is the purpose of an array in programming?
 - A. To store and manipulate text-based data
 - B. To define conditional statements
 - C. To organize and store multiple values of the same data type
 - D. To control the execution flow of a program
9. What is the term used to describe an error-free program that produces the expected results?
 - A. Syntax error
 - B. Runtime error
 - C. Logic error
 - D. Correct program
10. What does the term "variable scope" refer to in programming?
 - A. The physical memory location of a variable
 - B. The lifetime and accessibility of a variable
 - C. The process of assigning a value to a variable
 - D. The data type of a variable
11. Which type of error occurs during the execution of a program?
 - A. Syntax error
 - B. Runtime error
 - C. Logical error
 - D. Compile-time error
12. Which term refers to a named block of code that can be executed by calling its name?
 - A. Variable
 - B. Function
 - C. Class
 - D. Object
13. What is the purpose of conditional statements in computer programming?
 - A. To repeat a block of code multiple times
 - B. To define the structure of a program
 - C. To control the flow of program execution based on conditions
 - D. To perform mathematical calculations
14. In computer programming, what does the term "asymptotic analysis" refer to?
 - A. The process of debugging code
 - B. The study of algorithm efficiency as input sizes grow
 - C. The design of user interfaces
 - D. The practice of code refactoring
15. What is the main focus of the theory of computation?
 - A. To identify potential suppliers
 - B. To determine the optimal pricing strategy
 - C. To understand how consumers make purchasing decisions
 - D. To evaluate the performance of marketing campaigns
16. What is the primary goal of clean code development?
 - A. Writing code without any bugs or errors
 - B. Maximizing code efficiency and performance
 - C. Enhancing code readability and maintainability



- D. Minimizing code complexity and size
17. What does the SOLID principle "L" stand for?
- A. Least Privilege
 - B. Liskov Substitution
 - C. Low Coupling
 - D. Linting and Testing
18. Which of the following is a good practice for naming variables?
- A. Using single-letter variable names to save space
 - B. Using generic names like "data" or "temp"
 - C. Using descriptive and meaningful names
 - D. Using reserved keywords as variable names
19. What is the purpose of writing unit tests?
- A. To verify the correctness of the entire system
 - B. To identify performance bottlenecks in the code
 - C. To ensure that individual units of code work as expected
 - D. To generate code coverage reports
20. What is the purpose of code refactoring?
- A. To introduce new features into the codebase
 - B. To improve the performance of the code
 - C. To rewrite the entire code from scratch
 - D. To improve the structure and design of the code
21. Which software development methodology emphasizes frequent communication and collaboration among team members?
- A. Waterfall
 - B. Agile
 - C. Spiral
 - D. Extreme Programming
22. Which formal model of computation is based on the concept of a hypothetical machine with an infinite tape?
- A. Turing machine
 - B. Finite automaton
 - C. Pushdown automaton
 - D. Non-deterministic Turing machine
23. Which programming paradigm focuses on breaking down a problem into smaller, reusable functions?
- A. Object-oriented programming
 - B. Functional programming
 - C. Procedural programming
 - D. Event-driven programming
24. What is the role of comments in code?
- A. To prevent other developers from modifying the code
 - B. To provide additional information and context about the code
 - C. To generate API documentation automatically
 - D. To explain the basic syntax of the programming language
25. What is exception handling in programming?
- A. A technique to handle unexpected errors during program execution.
 - B. A method to enhance program performance.
 - C. A way to improve code readability.
 - D. A process to generate custom error messages.
26. What is the purpose of a layout manager in GUI programming?
- A. To handle user input events.
 - B. To define the visual appearance of the GUI components.
 - C. To manage the positioning and sizing of GUI components within a container.
- D. To handle data binding between GUI components and data sources.
27. What is an event-driven programming paradigm in the context of GUI programming?
- A. A programming style that focuses on code modularity and reusability.
 - B. A programming style that emphasizes strong typing and static analysis.
 - C. A programming style that relies on user actions or system events to trigger program behavior.
 - D. A programming style that utilizes graphical elements and visual design principles.
28. Which of the following is not a fundamental programming language construct?
- A. Variables
 - B. Functions
 - C. Loops
 - D. Classes
29. Which programming paradigm is based on the concept of "objects" and emphasizes encapsulation, inheritance, and polymorphism?
- A. Procedural programming
 - B. Functional programming
 - C. Object-oriented programming
 - D. Imperative programming
30. Which of the following is not a primitive data type in most programming languages?
- A. Integer
 - B. String
 - C. Boolean
 - D. Array
31. What is the purpose of conditional statements in programming?
- A. To handle errors and exceptions.
 - B. To define loops and iterations.
 - C. To control program flow based on certain conditions.



- D. To perform mathematical calculations.
32. What is the primary goal of structured programming?
- A. Efficient memory utilization
 - B. Simplicity and clarity of code
 - C. Object-oriented programming
 - D. High-level language usage
33. Which control structure is used in structured programming to perform repetitive tasks?
- A. If-else statement
 - B. Switch statement
 - C. For loop
 - D. While loop
34. What is the purpose of modular programming in structured programming?
- A. To improve code readability
 - B. To enhance program performance
 - C. To promote code reusability
 - D. To ensure error-free execution
35. Which concept in structured programming helps in breaking down a complex problem into smaller, manageable subproblems?
- A. Inheritance
 - B. Polymorphism
 - C. Abstraction
 - D. Decomposition
36. Which programming construct is used to selectively execute a block of code based on a condition?
- A. If-else statement
 - B. Switch statement
 - C. For loop
 - D. While loop
37. Which programming principle in structured programming emphasizes the avoidance of unrestricted use of jumps or GOTO statements?
- A. Encapsulation
 - B. Information hiding
 - C. Modularization
- D. Control flow
38. Which programming language feature is commonly used in structured programming to handle exceptions and errors?
- A. Functions
 - B. Classes
 - C. Pointers
 - D. Exception handling
39. Which of the following is a key feature of structured programming?
- A. Object-oriented programming
 - B. Use of GOTO statements
 - C. Use of subroutines
 - D. No use of conditionals
40. Which of the following is NOT a control structure used in structured programming?
- A. Recursion
 - B. Sequence
 - C. Iteration
 - D. Selection
41. Write a pseudocode to calculate the sum of the first 100 natural numbers.
- A. for i = 1 to 100: sum = sum + i
 - B. sum = 100 * (100 + 1) / 2
 - C. sum = 100 * 100 / 2
 - D. sum = 100 * (100 - 1) / 2
42. Write a pseudocode to find the maximum of three numbers: a, b, and c.
- A. max = a if a > b and a > c else b if b > c else c
 - B. max = a if a > b else b if b > c else c
 - C. max = a if a > b else b if b > c and a > c else c
 - D. max = a if a > b and a > c else b if b > c and a < c else c
43. Write a pseudocode to check if a given number is even or odd.
- A. if num % 2 == 0: print "Even" else: print "Odd"
- B. if num % 2 == 0: output "Even" else: output "Odd"
- C. if num % 2 == 0: return "Even" else: return "Odd"
- D. if num % 2 == 0: print "Odd" else: print "Even"
44. Write a pseudocode to calculate the factorial of a given number.
- A. factorial = 1
for i = 1 to n:
factorial = factorial * i
endfor
 - B. factorial = n
while n > 1:
factorial = factorial * (n - 1)
n = n - 1
endwhile
 - C. factorial = 1
for i = n to 1:
factorial = factorial * i
endfor
 - D. factorial = n
while n > 1:
factorial = factorial * n
n = n - 1
endwhile
45. Write a pseudocode to find the sum of the first n even numbers.
- A. sum = 0
for i = 1 to n:
sum = sum + (2 * i)
endfor
 - B. sum = 0
i = 2 to n * 2 step 2:
sum = sum + i
endfor
 - C. sum = 0
for i = 1 to n * 2:
sum = sum + i
endfor
 - D. sum = 0
i = 2 to n:
sum = sum + (2 * i)
endfor
46. Write a pseudocode to check if a given number is a prime number.
- A. is_prime = true
for i = 2 to num - 1:
if num % i == 0:
is_prime = false
exit loop



- endif
endfor
- B. is_prime = true
for i = 2 to num / 2:
if num % i == 0:
is_prime = false
exit loop
endif
endfor
- C. is_prime = true
for i = 2 to sqrt(num):
if num % i == 0:
is_prime = false
exit loop
endif
endfor
- D. is_prime = true
for i = 2 to num:
if num % i == 0:
is_prime = false
exit loop
endif
endfor
47. Write a pseudocode to calculate the Fibonacci sequence up to the nth term
- A. fibonacci[0] = 0
fibonacci[1] = 1
for i = 2 to n:
fibonacci[i] = fibonacci[i-1] + fibonacci[i-2]
endfor
- B. fibonacci[0] = 0
fibonacci[1] = 1
for i = 2 to n:
fibonacci[i] = fibonacci[i-2] + fibonacci[i-1]
endfor
- C. fibonacci[0] = 0
fibonacci[1] = 1
for i = 2 to n:
fibonacci[i] = fibonacci[i-1] * fibonacci[i-2]
endfor
- D. fibonacci[0] = 0
fibonacci[1] = 1
for i = 2 to n:
fibonacci[i] = fibonacci[i-1] - fibonacci[i-2]
endfor
48. Write a pseudocode to calculate the sum of all even numbers between 1 and 100 (inclusive).
- A. sum = 0
for i = 1 to 100:
if i % 2 == 0:
sum = sum + i
endif
endfor
- B. sum = 0
for i = 2 to 100 step 2:
sum = sum + i
endfor
- C. sum = 0
for i = 1 to 100 step 2:
sum = sum + i
endfor
- D. sum = 0
for i = 2 to 100:
if i % 2 == 0:
sum = sum + i
endif
endfor
49. Write a pseudocode to count the number of vowels in a given string.
- A. vowels = 0
for i = 0 to length(string) - 1:
if string[i] is a vowel:
vowels = vowels + 1
endif
endfor
- B. vowels = 0
for i = 0 to length(string) - 1:
if string[i] in ["a", "e", "i", "o", "u"]:
vowels = vowels + 1
endif
endfor
- C. vowels = 0
for i = 0 to length(string):
if string[i] is a vowel:
vowels = vowels + 1
endif
endfor
- D. vowels = 0
for i = 0 to length(string):
if string[i] in ["a", "e", "i", "o", "u"]:
vowels = vowels + 1
endif
endfor
50. Which of the following statements about pseudocode is true?
- A. Pseudocode is a programming language.
- B. Pseudocode is executed by a computer.
- C. Pseudocode is used to write real, executable code.
- D. Pseudocode is a way to represent algorithms in a human-readable form.